C. James Nesbit IV (CJ)

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ABOUT

Chemical Engineering student at the University of Dayton with a passion for advancing energy systems and solutions. Skilled in data analysis, computer programming, and software literacy. Proven ability to deliver accurate, high-quality results through effective team collaboration. Actively seeking opportunities to apply technical skills to solve real-world energy and engineering problems.

EDUCATION

University of Dayton, Dayton OH — May 2027
Bachelor of Engineering, Chemical Engineering
GPA: 4.0/4.0

WORK EXPERIENCE

Marathon Petroleum Corporation, Findlay OH — May 2026 – August 2026
Operations Research Co-op

- Developed real-time calculations for heat transfer coefficients on lean/rich solvent heat exchangers to monitor long-term performance.
- Created and migrated 20+ new Key Performance Indicator (KPI) calculations for refinery HF Alky units. Communicated with multiple unit engineers across sites to ensure data quality.
- Validated 50+ Focus On Energy (FOE) metric calculations across reboiler, steam generator, and stripping steam equipment to improve energy efficiency efforts.

Marathon Petroleum Corporation, Robinson IL — January 2026 – May 2026
Refining Engineering Co-op

- Used process modeling software and engineering calculations to update a water wash procedure to conform to refinery standards. Summarized findings and recommendations in a technical memo.
- Created inspection checklists for 10+ individual vessels to support unit engineer responsibilities during turnaround (TAR).
- Developed predictive Excel tool to automatically anticipate tower flooding based on historical pressure drop in an isomerization stabilizer unit.

University of Dayton Research Institute, Dayton OH — October 2024 – December 2025
Fuel Science Intern

- Analyzed gas chromatography (GC-FID, GCxGC) samples for 80+ international jet fuels.
- Researched the effects of fuel additives on dielectric constant measurements used for fuel gauging. Led 20+ individual experiments with unique fuel & additive combinations to develop models to accurately predict the magnitude of additive effects.
- Performed 50+ surface tension experiments with pure components of standard jet fuels to commission new equipment. Researched effects of 5 various needle diameters, temperature and air flow variables.

University of Dayton Research Institute, Dayton OH — January 2025 – August 2025
Chemical Process Co-op

- Assisted with the development of a mobile petroleum desulfurization unit for the US Air Force.
- Gained hands-on experience with the design and construction of various refining process units. Gained experience reading and developing Process Flow Diagrams and Piping & Instrumentation drawings.
- Optimized mechanical bill of materials and organized suppliers to save $11,000.

KEY SKILLS

- Aspen Plus, Aspen HYSYS, Process Modeling, Process Control, Process Engineering
- Software Literacy, C++, Java, HTML, CSS, VBA, Microsoft Office, Adobe Creative Suite
- Communication, Team Collaboration, Project Management, Public Speaking, Problem Solving